

ADDENDUM 4 to KIMI_LPW_REVIEW_VERDICT.md — custody convention and prose corrections (2026-09-13)

The sealed verdict and Addenda 1–3 remain unchanged. This fourth addendum resolves the custody convention found by the Return-05 archive audit (zip sha256 ae03fbbd..., manifest verified) and corrects three prose items. The LPW endorsement (cf58f72e...e1a5) is untouched.

1. The six-byte body-hash convention — resolved at the exact byte range

The verdict’s declared body hash d399e28378d177f88d08032c81f1573f35c9af2b600b9b775ebfe40f555cbbad covers the FIRST 9,084 BYTES of KIMI_LPW_REVIEW_VERDICT.md: everything before the separator block

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"\n---\n\n" + "SHA-256 of this report body (text after this line is excluded):\n" + hex + "\n"
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— i.e., the body is the file’s content up to but excluding the “—” separator that precedes the marker line. Verified on the delivered bytes: sha256(first 9,084 bytes) = d399e28378d177f8... (matches the declaration exactly). The README’s earlier phrasing (“hash all bytes before the marker line”) covers 9,090 bytes (including the 6-byte separator “—”) and hashes to 17863be2e9fedc4d... — that phrasing was imprecise, not the hash. GOVERNING CONVENTION (bound here for all KIMI carriers): the body-hash marker block is the separator “—” followed by the marker line and one hex line; the declared body hash is the SHA-256 of everything BEFORE that block; the whole-file hash additionally covers the complete file including the block. No bytes are trimmed or reserialized on either side of the rule.

2. Cov(J) off-diagonal prose — corrected

My lead review of LPW_CONSTANT said the off-diagonal entries of Cov(J) “vanish by parity at $t = 0$ ”. That is wrong for the pair $(f_{xxy}/2, f_{yyy}/6)$: $\text{Cov}(f_{xxy}/2, f_{yyy}/6) = a a / 12 \neq 0$ (per-axis orders (2,1) vs (0,3): even in both coordinates). The code’s Gershgorin calculation includes the term correctly: row sums are $3(f_{yy})$, $a a / 4 + a a / 12 = 1.0(f_{xxy}/2)$, $1.0(f_{xxy}/2)$, and $a a / 12 + a / 36 = 0.667(f_{yyy}/6)$, so $\Lambda = \max \text{row sum} = 3$ UNAFFECTED. Only the prose claim was wrong; no number in the chain changes. The conditional endpoint Schur complement (diagonal at the planar reference) is a different object from the unconditioned Cov(J) and is unaffected.

3. The fallback’s domain — corrected

My LPW_CONSTANT review said the author-side fallback $(r_* = 10^{-28})$ “extends the tiny-r regime”. That is wrong as stated: $(0, 10^{-28}] \cap (0, 1/2414592]$, so the certified pair already covers every radius the fallback names. The fallback’s value is as a SEPARATELY JUSTIFIED ROUTE (its own analytical chain, useful for cross-checking and for exposing which estimates dominate), not as additional radius coverage.

4. Old eigenvalue wording — final note

Every floor in the campaign now rounds DOWNWARD (the proved conservative endpoint floor 0.124 with the value 0.126553449667289 as evidence). The earlier upward-rounded display is withdrawn everywhere and must not be revived in any successor prose.

5. The Return-05 defects — handled in the corrected successor (v2)

The two LPW_CONSTANT defects the audit found are repaired in LPW_CONSTANT/v2/lpw_constant_v2.py (new version; v1 frozen): the published decimal now derives from the exact integer denominator ($c = 260/(B_{3,\text{rat}} \cdot 2^{21})$) with $B_{3,\text{rat}} = 2082000000000$ giving $1.135776\text{e-}43$, published downward-safe at $1.13\text{e-}43$ and floored at 10^{-43}), and the false fourth-moment identity is replaced by the author-side amplitude majorant ($E[R] = 8$, $E[R] \leq \sqrt{2}$), whose multiplier is smaller than the false one — v1's caps survive conservatively, and v2 publishes the improved pair the repair yields ($c = 1.13\text{e-}43$, $r = 1/2224640$). See LPW_CONSTANT/v2/LPW_CONSTANT_V2_REPORT.md.

SHA-256 of this addendum body (text after this line is excluded): 0f0e607c2d0accf357c48ce4a0ea760924436373d408475890c